Chapter 9. Experiments with Random Factors

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1 Introduction: Fixed vs Random Factors

- **Fixed factor**: a specific set of levels is chosen; inference is confined to those levels. Quantitative factors are usually fixed (the levels are deliberately chosen to span a region, and one wants a response curve over exactly that region).
- **Random factor**: levels are chosen at random from a larger population of potential levels; inference is about the *entire population* of levels. Industrial applications include measurement-system (gauge R&R) studies.

2 The One-Way Random Effects Model

2.1 Example: looms

A textile company weaves fabric on a large number (hundreds) of looms; is there a difference in strength due to looms? Four looms are selected *at random* and four strength determinations are made on each ($a = 4$, $n = 4$). Looms are a random factor.

Loom	Observations				$y_{i\cdot}$
1	98	97	99	96	390
2	91	90	93	92	366
3	96	95	97	95	383
4	95	96	99	98	388
					1527

2.2 Model and covariance structure

Definition 9.1

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, n,$$

$\tau_i \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2)$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, $\tau_i \perp\!\!\!\perp \varepsilon_{ij}$. σ_τ^2 and σ^2 are **variance components**.

Derivation

$$\begin{aligned}
\mathrm{E}(y_{ij}) &= \mu, \qquad \mathrm{Var}(y_{ij}) = \mathrm{Var}(\tau_i) + \mathrm{Var}(\varepsilon_{ij}) = \sigma_\tau^2 + \sigma^2, \\
\mathrm{Cov}(y_{ij}, y_{ij'}) &= \mathrm{Cov}(\tau_i + \varepsilon_{ij}, \tau_i + \varepsilon_{ij'}) = \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_i + \varepsilon_{ij'})\big] = \mathrm{E}(\tau_i^2) = \sigma_\tau^2 \quad (j \neq j'), \\
\mathrm{Cov}(y_{ij}, y_{i'j'}) &= \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_{i'} + \varepsilon_{i'j'})\big] = 0 \quad (i \neq i').
\end{aligned}$$

So with $\mathbf{y} = (y_{11}, \dots, y_{1n}, \dots, y_{a1}, \dots, y_{an})^T$,

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{y}) = \begin{pmatrix} \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n & & \\ & \ddots & \\ & & \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n \end{pmatrix} = \mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n),$$

a block-diagonal matrix: observations from the same loom are correlated (intraclass correlation $\rho = \sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$), different looms are independent. Under $H_0 : \sigma_\tau^2 = 0$, $\mathrm{Cov}(\mathbf{y}) = \sigma^2\mathbf{I}$.

2.3 Hypotheses

The individual treatments are not of specific interest; we care about the population:

$H_0 : \sigma_\tau^2 = 0$ (all treatments identical) vs $H_1 : \sigma_\tau^2 > 0$ (variability exists between treatments).

2.4 ANOVA and expected mean squares

The partition $SS_T = SS_{\text{Treatments}} + SS_E$ with df $an - 1 = (a-1) + a(n-1)$ is purely algebraic and still holds.

Theorem 9.2

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + n\sigma_\tau^2, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ and $\bar{y}_{\cdot\cdot} = \mu + \bar{\tau} + \bar{\varepsilon}_{\cdot\cdot}$ (now $\bar{\tau} = \frac{1}{a}\sum \tau_i$ is random, not 0). Thus

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = (\tau_i - \bar{\tau}) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}),$$

and the two brackets are independent with mean zero, so the cross term has expectation zero:

$$\mathrm{E}(MS_{\text{Treatments}}) = \mathrm{E}\Big[\frac{n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}{a-1}\Big] = n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\tau_i - \bar{\tau})^2}{a-1}\Big]}_{=\mathrm{Var}(\tau_i)=\sigma_\tau^2} + n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot})=\sigma^2/n} = n\sigma_\tau^2 + \sigma^2.$$

(A sample variance of iid variables is unbiased for their variance.) For the error, $y_{ij} - \bar{y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$ (τ_i cancels), so

$$\mathrm{E}(MS_E) = \mathrm{E}\Big[\frac{\sum\sum(y_{ij} - \bar{y}_{i\cdot})^2}{a(n-1)}\Big] = \frac{1}{a}\sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{n-1}\Big] = \sigma^2.$$

■

Test: $F_0 = MS_{\text{Treatments}}/MS_E \overset{H_0}{\sim} F_{a-1,N-a}$; reject $H_0 : \sigma_\tau^2 = 0$ if $F_0 > F_{\alpha,a-1,N-a}$. (Same computation as the fixed case; only the hypothesis and interpretation differ.)

2.5 Quadratic-form treatment

Theorem 9.3

Let $\mathbf{A} = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n - \frac{1}{N}\mathbf{J}_N$. Then (a) $SS_{\text{Treatments}} = \mathbf{y}^T\mathbf{A}\mathbf{y}$;
(b) $\mathbf{A}\boldsymbol{\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$, so $\mathbf{A}\boldsymbol{\Sigma}/(\sigma^2 + n\sigma_\tau^2) = \mathbf{A}$ is idempotent and $SS_{\text{Treatments}}/(\sigma^2 + n\sigma_\tau^2) \sim \chi^2_{a-1}$;
(c) $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = (a-1)(\sigma^2 + n\sigma_\tau^2)$.

Proof [Supplement]. (a) $\frac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)\mathbf{y}$ replaces each y_{ij} by $\bar{y}_{i\cdot}$, and $\frac{1}{N}\mathbf{J}_N\mathbf{y} = \bar{y}_{\cdot\cdot}\mathbf{1}$. So $\mathbf{A}\mathbf{y}$ is the vector of $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$. $\mathbf{A}$ is symmetric and idempotent (it is $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ of Ch. 3), hence $\mathbf{y}^T\mathbf{A}\mathbf{y} = (\mathbf{A}\mathbf{y})^T(\mathbf{A}\mathbf{y}) = \sum_i\sum_j(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

(b) Using $(\mathbf{P}\otimes\mathbf{Q})(\mathbf{R}\otimes\mathbf{S}) = \mathbf{PR}\otimes\mathbf{QS}$ and $\mathbf{J}_n\mathbf{J}_n = n\mathbf{J}_n$:

$$\tfrac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \mathbf{I}_a \otimes \tfrac{1}{n}(\sigma^2\mathbf{J}_n + n\sigma_\tau^2\mathbf{J}_n) = (\sigma^2 + n\sigma_\tau^2)\,\tfrac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n,$$
$$\tfrac{1}{N}(\mathbf{J}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \tfrac{1}{N}\mathbf{J}_a \otimes (\sigma^2 + n\sigma_\tau^2)\mathbf{J}_n = (\sigma^2 + n\sigma_\tau^2)\tfrac{1}{N}\mathbf{J}_N.$$

Subtracting, $\mathbf{A\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$. For $\mathbf{y} \sim N(\mu\mathbf{1}, \mathbf{\Sigma})$, $\mathbf{y}^T\mathbf{B}\mathbf{y} \sim \chi^2_{\text{rank}\mathbf{B}}(\cdot)$ when $\mathbf{B\Sigma}$ is idempotent; with $\mathbf{B} = \mathbf{A}/(\sigma^2 + n\sigma_\tau^2)$, $\mathbf{B\Sigma} = \mathbf{A}$ is idempotent of rank $\text{tr}\mathbf{A} = a - 1$, and the noncentrality $\mu^2\mathbf{1}^T\mathbf{B}\mathbf{1} = 0$ since $\mathbf{A}\mathbf{1} = \mathbf{0}$.

(c) $\text{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \text{tr}(\mathbf{A\Sigma}) + \mu^2\mathbf{1}^T\mathbf{A}\mathbf{1} = (\sigma^2 + n\sigma_\tau^2)\text{tr}(\mathbf{A}) + 0 = (a-1)(\sigma^2 + n\sigma_\tau^2)$. ■

2.6 Estimating variance components

ANOVA (moment) estimators, from equating MS to EMS:

$$\hat{\sigma}^2 = MS_E, \qquad \hat{\sigma}_\tau^2 = \frac{MS_{\text{Treatments}} - MS_E}{n}, \qquad \widehat{\text{Var}}(y_{ij}) = \hat{\sigma}_y^2 = \hat{\sigma}_\tau^2 + \hat{\sigma}^2.$$

Unbiased: $\text{E}(\hat{\sigma}_\tau^2) = \frac{(\sigma^2 + n\sigma_\tau^2) - \sigma^2}{n} = \sigma_\tau^2$. Also $\hat{\sigma}_\tau^2 = \widehat{\text{Cov}}(y_{ij}, y_{ij'})$ and $\widehat{\text{Cov}}(\mathbf{y})$ has $\hat{\sigma}^2 + \hat{\sigma}_\tau^2$ on the diagonal blocks and $\hat{\sigma}_\tau^2$ off-diagonal within blocks.

Potential problems:

(1) **negative estimates** when $MS_{\text{Treatments}} < MS_E$;

(2) they are moment estimators and do not have the best statistical properties (REML/ML via `lme` are alternatives).

2.7 Confidence intervals

For σ^2. Under normality $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$, so

$$P\Big(\chi^2_{1-\alpha/2,N-a} \le \frac{(N-a)MS_E}{\sigma^2} \le \chi^2_{\alpha/2,N-a}\Big) = 1 - \alpha \;\Rightarrow\; \frac{(N-a)MS_E}{\chi^2_{\alpha/2,N-a}} \le \sigma^2 \le \frac{(N-a)MS_E}{\chi^2_{1-\alpha/2,N-a}}.$$

For σ_τ^2: no exact interval exists.

For the intraclass correlation $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$. Since $\frac{(a-1)MS_{\text{Treatments}}}{\sigma^2 + n\sigma_\tau^2} \sim \chi^2_{a-1}$ and is independent of $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$,

$$\mathcal{A} = \frac{MS_{\text{Treatments}}/(n\sigma_\tau^2 + \sigma^2)}{MS_E/\sigma^2} \sim F_{a-1,N-a} \quad \Rightarrow \quad P(F_{1-\alpha/2,a-1,N-a} \le \mathcal{A} \le F_{\alpha/2,a-1,N-a}) = 1 - \alpha.$$

Inverting the pivot [Supplement]. Let $\theta = \sigma_\tau^2/\sigma^2$ and $F^* = MS_{\text{Treatments}}/MS_E$. Then $\mathcal{A} = F^*/(1 + n\theta)$ and

$$F_{1-\alpha/2} \le \frac{F^*}{1+n\theta} \le F_{\alpha/2} \iff \frac{1}{n}\Big(\frac{F^*}{F_{\alpha/2}} - 1\Big) \le \theta \le \frac{1}{n}\Big(\frac{F^*}{F_{1-\alpha/2}} - 1\Big).$$

Call these bounds L and U. Since $\frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} = \frac{\theta}{1+\theta}$ is increasing in θ,

$$\frac{L}{1+L} \le \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} \le \frac{U}{1+U},$$

$$L = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{\alpha/2,a-1,N-a}} - 1\Big), \quad U = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{1-\alpha/2,a-1,N-a}} - 1\Big).$$

■

Example 9.4: Looms, line by line

$\sum y^2 = 145845$, $y_{\cdot\cdot}^2/N = 1527^2/16 = 145733.06$.

$$SS_T = 145845 - 145733.06 = 111.94,$$
$$SS_{\text{Looms}} = \frac{390^2 + 366^2 + 383^2 + 388^2}{4} - 145733.06 = \frac{583289}{4} - 145733.06 = 89.19,$$
$$SS_E = 111.94 - 89.19 = 22.75.$$

Source	SS	DF	MS	F_0
Looms	89.19	3	29.73	15.68
Error	22.75	12	1.90	
Total	111.94	15		

$F_0 = 15.68 > F_{0.05,3,12} = 3.49$ ($P = 0.00019$): reject $H_0 : \sigma_\tau^2 = 0$.

$$\hat\sigma^2 = 1.90, \qquad \hat\sigma_\tau^2 = \frac{29.73 - 1.90}{4} = 6.96, \qquad \hat\sigma_y^2 = 6.96 + 1.90 = 8.86$$

(`lme`: 6.958 and 1.896). Most of the variability ($6.96/8.86 = 78.6\%$) is due to looms.
95% CI for σ^2: $\chi^2_{0.025,12} = 23.34$, $\chi^2_{0.975,12} = 4.40$: $[\frac{12(1.896)}{23.34}, \frac{12(1.896)}{4.40}] = [0.975,\ 5.17]$.
95% CI for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$: $F_{0.025,3,12} = 4.47$, $F_{0.975,3,12} = 0.0698$, $F^* = 15.68$:

$$L = \tfrac{1}{4}(\tfrac{15.68}{4.47} - 1) = 0.626, \quad U = \tfrac{1}{4}(\tfrac{15.68}{0.0698} - 1) = 55.95, \quad \left[\tfrac{0.626}{1.626}, \tfrac{55.95}{56.95}\right] = [0.385,\ 0.982].$$

3 Two-Factor Factorial with Random Factors

Motivation. When the levels of a factor are a random sample from a larger population (operators, batches), we care about the variance they contribute, not their individual effects. Random-effects models change the expected mean squares and therefore which ratio forms the correct F-test.

Definition 9.5

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\tau_i \sim N(0, \sigma_\tau^2)$, $\beta_j \sim N(0, \sigma_\beta^2)$, $(\tau\beta)_{ij} \sim N(0, \sigma_{\tau\beta}^2)$, $\varepsilon_{ijk} \sim N(0, \sigma^2)$, all independent.
$\text{Var}(y_{ijk}) = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{\tau\beta}^2 + \sigma^2$.

Hypotheses: $H_0 : \sigma_\tau^2 = 0$, $H_0 : \sigma_\beta^2 = 0$, $H_0 : \sigma_{\tau\beta}^2 = 0$. The ANOVA computations are unchanged; the tests change.

Theorem 9.6

$$\text{E}(MS_A) = \sigma^2 + n\sigma_{\tau\beta}^2 + bn\sigma_\tau^2, \quad \text{E}(MS_B) = \sigma^2 + n\sigma_{\tau\beta}^2 + an\sigma_\beta^2,$$
$$\text{E}(MS_{AB}) = \sigma^2 + n\sigma_{\tau\beta}^2, \quad \text{E}(MS_E) = \sigma^2.$$

$\text{E}(MS_A)$ **[Supplement].**

$$\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta + \overline{(\tau\beta)}_{i\cdot} + \bar\varepsilon_{i\cdot\cdot}, \qquad \bar y_{\cdots} = \mu + \bar\tau + \bar\beta + \overline{(\tau\beta)}_{\cdot\cdot} + \bar\varepsilon_{\cdots},$$
$$SS_A = bn \sum_i \left[(\tau_i - \bar\tau) + (\overline{(\tau\beta)}_{i\cdot} - \overline{(\tau\beta)}_{\cdot\cdot}) + (\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdots})\right]^2.$$

$\bar\beta$ cancels. The three brackets are independent with mean 0, so cross terms vanish in expectation, and each is a sample-variance-type sum of iid terms:

$$\mathrm{E}(MS_A) = bn\big[\mathrm{Var}(\tau_i) + \mathrm{Var}(\overline{(\tau\beta)}_{i\cdot}) + \mathrm{Var}(\bar\varepsilon_{i\cdot\cdot})\big] = bn\Big(\sigma^2_\tau + \frac{\sigma^2_{\tau\beta}}{b} + \frac{\sigma^2}{bn}\Big) = bn\sigma^2_\tau + n\sigma^2_{\tau\beta} + \sigma^2.$$

By symmetry $\mathrm{E}(MS_B) = an\sigma^2_\beta + n\sigma^2_{\tau\beta} + \sigma^2$. For MS_{AB}: $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdots}$ removes τ_i, β_j and leaves the RCBD-type residual of the table $\{(\tau\beta)_{ij} + \bar\varepsilon_{ij\cdot}\}$ of iid variables with variance $\sigma^2_{\tau\beta} + \sigma^2/n$; its expected sum of squares is $(a-1)(b-1)(\sigma^2_{\tau\beta} + \sigma^2/n)$, hence $\mathrm{E}(MS_{AB}) = n\sigma^2_{\tau\beta} + \sigma^2$. ■

Tests (choose the denominator whose EMS equals the numerator's under H_0):

$$H_0 : \sigma^2_\tau = 0 : \ F_0 = \frac{MS_A}{MS_{AB}}, \qquad H_0 : \sigma^2_\beta = 0 : \ F_0 = \frac{MS_B}{MS_{AB}}, \qquad H_0 : \sigma^2_{\tau\beta} = 0 : \ F_0 = \frac{MS_{AB}}{MS_E}.$$

Estimators:

$$\hat\sigma^2_\tau = \frac{MS_A - MS_{AB}}{bn}, \quad \hat\sigma^2_\beta = \frac{MS_B - MS_{AB}}{an}, \quad \hat\sigma^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n}, \quad \hat\sigma^2 = MS_E.$$

Example 9.7: Gauge capability study

20 parts (random) × 3 operators (random), $n = 2$ measurements each.

Source	SS	DF	MS	F_0	denominator
Parts	1185.43	19	62.39	$62.39/0.71 = 87.9$	MS_{AB}
Operators	2.62	2	1.31	$1.31/0.71 = 1.84$	MS_{AB}
P×O	27.05	38	0.71	$0.71/0.99 = 0.72$	MS_E
Error	59.50	60	0.99		

$$\hat\sigma^2_\tau = \frac{62.39 - 0.71}{3\cdot 2} = 10.28, \quad \hat\sigma^2_\beta = \frac{1.31 - 0.71}{20\cdot 2} = 0.015,$$
$$\hat\sigma^2_{\tau\beta} = \frac{0.71 - 0.99}{2} = -0.14, \quad \hat\sigma^2 = 0.99.$$

The negative $\hat\sigma^2_{\tau\beta}$ is consistent with a zero interaction component; one may refit the reduced model $y = \mu + \tau_i + \beta_j + \varepsilon$. Gauge variability $\hat\sigma^2_{\text{gauge}} = \hat\sigma^2 + \hat\sigma^2_\beta$ (repeatability + reproducibility) is small relative to part variability.

4 Two-Factor Mixed Models

Motivation. With one fixed and one random factor the correct denominator for testing the fixed factor is the interaction mean square, not the error. The expected mean squares derived here decide this.

4.1 Restricted mixed model

Definition 9.8: A fixed, B random (restricted model)

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\sum_i \tau_i = 0$; $\beta_j \overset{\text{iid}}{\sim} N(0, \sigma_\beta^2)$; $(\tau\beta)_{ij} \sim N\big(0, \frac{a-1}{a}\sigma_{\tau\beta}^2\big)$ with the restriction $\sum_{i=1}^{a}(\tau\beta)_{ij} = (\tau\beta)_{\cdot j} = 0$ for each j; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

Interaction terms in the same column are therefore *not* independent:

Theorem 9.9

$\operatorname{Cov}\big[(\tau\beta)_{ij}, (\tau\beta)_{i'j}\big] = -\frac{1}{a}\sigma_{\tau\beta}^2$ $(i \neq i')$, and $\operatorname{Cov}\big[(\tau\beta)_{ij}, (\tau\beta)_{ij'}\big] = 0$ $(j \neq j')$.

Proof [Supplement]. Let $x = \operatorname{Cov}[(\tau\beta)_{ij}, (\tau\beta)_{i'j}]$ (same for all $i \neq i'$ by symmetry). Since $\sum_i (\tau\beta)_{ij} = 0$,

$$0 = \operatorname{Var}\Big(\sum_{i=1}^{a}(\tau\beta)_{ij}\Big) = \sum_{i=1}^{a}\operatorname{Var}\big((\tau\beta)_{ij}\big) + 2\binom{a}{2}x = a\cdot\frac{a-1}{a}\sigma_{\tau\beta}^2 + a(a-1)x \;\Rightarrow\; x = -\frac{\sigma_{\tau\beta}^2}{a}.$$

Different columns j correspond to independently drawn random levels, so their interactions are uncorrelated. ∎

Theorem 9.10: EMS, restricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma_{\tau\beta}^2 + \frac{bn\sum_i \tau_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + an\sigma_\beta^2,$$
$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma_{\tau\beta}^2, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. MS_A. $\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta + \overline{(\tau\beta)}_{i\cdot} + \bar\varepsilon_{i\cdot\cdot}$; $\bar y_{\cdots} = \mu + \bar\tau + \bar\beta + \overline{(\tau\beta)}_{\cdot\cdot} + \bar\varepsilon_{\cdots}$ with $\bar\tau = 0$ and $\overline{(\tau\beta)}_{\cdot\cdot} = 0$ (restriction). So $\bar y_{i\cdot\cdot} - \bar y_{\cdots} = \tau_i + \overline{(\tau\beta)}_{i\cdot} + (\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdots})$ and

$$\mathrm{E}(MS_A) = bn\frac{\sum\tau_i^2}{a-1} + \underbrace{bn\,\mathrm{E}\Big(\frac{\sum_i \overline{(\tau\beta)}_{i\cdot}^{\,2}}{a-1}\Big)}_{(*)} + \underbrace{bn\,\mathrm{E}\Big(\frac{\sum_i(\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdots})^2}{a-1}\Big)}_{=\sigma^2/(bn)}.$$

$(*)$: $\overline{(\tau\beta)}_{i\cdot} = \frac{1}{b}\sum_j (\tau\beta)_{ij}$ is an average of b independent terms (different j), so

$$\mathrm{E}\Big(\sum_{i=1}^{a}\overline{(\tau\beta)}_{i\cdot}^{\,2}\Big) = \sum_i \operatorname{Var}\big(\overline{(\tau\beta)}_{i\cdot}\big) = \sum_{i=1}^{a}\frac{\operatorname{Var}((\tau\beta)_{ij})}{b} = \frac{a}{b}\cdot\frac{a-1}{a}\sigma_{\tau\beta}^2 = \frac{a-1}{b}\sigma_{\tau\beta}^2,$$

so $bn(*) = bn\cdot\frac{\sigma_{\tau\beta}^2}{b} = n\sigma_{\tau\beta}^2$. Hence $\mathrm{E}(MS_A) = \sigma^2 + n\sigma_{\tau\beta}^2 + \frac{bn\sum\tau_i^2}{a-1}$.

MS_B. $\bar y_{\cdot j\cdot} = \mu + \bar\tau + \beta_j + \overline{(\tau\beta)}_{\cdot j} + \bar\varepsilon_{\cdot j\cdot} = \mu + \beta_j + \bar\varepsilon_{\cdot j\cdot}$ because $\bar\tau = 0$ and $\overline{(\tau\beta)}_{\cdot j} = 0$ by the restriction. Thus $\bar y_{\cdot j\cdot} - \bar y_{\cdots} = (\beta_j - \bar\beta) + (\bar\varepsilon_{\cdot j\cdot} - \bar\varepsilon_{\cdots})$ and

$$\mathrm{E}(MS_B) = an\,\mathrm{E}\Big(\frac{\sum_j(\beta_j - \bar\beta)^2}{b-1}\Big) + an\,\mathrm{E}\Big(\frac{\sum_j(\bar\varepsilon_{\cdot j\cdot} - \bar\varepsilon_{\cdots})^2}{b-1}\Big) = an\sigma_\beta^2 + an\frac{\sigma^2}{an} = \sigma^2 + an\sigma_\beta^2.$$

The interaction does *not* appear in $\mathrm{E}(MS_B)$ in the restricted model.

MS_{AB}. $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdots} = \big((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot}\big) + (\varepsilon\text{-part})$. For fixed i, $\{(\tau\beta)_{ij}\}_{j=1}^{b}$ are iid with variance $\frac{a-1}{a}\sigma_{\tau\beta}^2$, so $\mathrm{E}\sum_j((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot})^2 = (b-1)\frac{a-1}{a}\sigma_{\tau\beta}^2$; summing over i: $(a-1)(b-1)\sigma_{\tau\beta}^2$. With the error part contributing $(a-1)(b-1)\sigma^2/n$, $\mathrm{E}(MS_{AB}) = n\sigma_{\tau\beta}^2 + \sigma^2$. ∎

Tests: $H_0 : \tau_i = 0$: $F_0 = MS_A/MS_{AB}$; $H_0 : \sigma_\beta^2 = 0$: $F_0 = MS_B/MS_E$; $H_0 : \sigma_{\tau\beta}^2 = 0$: $F_0 = MS_{AB}/MS_E$.
Estimators: $\hat\sigma_\beta^2 = \frac{MS_B - MS_E}{an}$, $\hat\sigma_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n}$, $\hat\sigma^2 = MS_E$, $\hat\tau_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}$.

4.2 Unrestricted mixed model

Definition 9.11

$$y_{ijk} = \mu + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + \varepsilon_{ijk},$$
$$\sum_i \alpha_i = 0, \quad \gamma_j \sim N(0, \sigma^2_\gamma), \quad (\alpha\gamma)_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2_{\alpha\gamma}), \quad \varepsilon_{ijk} \sim N(0, \sigma^2),$$

with *no* sum-to-zero restriction on the interactions (all mutually independent).

Theorem 9.12: EMS, unrestricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\alpha\gamma} + \frac{bn\sum \alpha_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + n\sigma^2_{\alpha\gamma} + an\sigma^2_\gamma,$$
$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\alpha\gamma}, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \alpha_i + [\overline{(\alpha\gamma)}_{i\cdot} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ ($\bar{\gamma}$ cancels, $\bar{\alpha} = 0$). The $\overline{(\alpha\gamma)}_{i\cdot}$ are iid with variance $\sigma^2_{\alpha\gamma}/b$, so

$$\mathrm{E}(MS_A) = \frac{bn\sum \alpha_i^2}{a-1} + bn \cdot \frac{\sigma^2_{\alpha\gamma}}{b} + bn \cdot \frac{\sigma^2}{bn} = \frac{bn\sum \alpha_i^2}{a-1} + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

$\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\gamma_j - \bar{\gamma}) + [\overline{(\alpha\gamma)}_{\cdot j} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ with $\mathrm{Var}(\overline{(\alpha\gamma)}_{\cdot j}) = \sigma^2_{\alpha\gamma}/a$, so

$$\mathrm{E}(MS_B) = an\sigma^2_\gamma + an \cdot \frac{\sigma^2_{\alpha\gamma}}{a} + an \cdot \frac{\sigma^2}{an} = an\sigma^2_\gamma + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

■

Consequently, in the unrestricted model $H_0 : \sigma^2_\gamma = 0$ is tested by $F_0 = MS_B/MS_{AB}$ and $\hat{\sigma}^2_\gamma = \frac{MS_B - MS_{AB}}{an}$. (Restricted vs unrestricted differ only in the test/estimator for the random main effect.)

5 Rules for Expected Mean Squares

1. The error term $\varepsilon_{ijk\ldots m}$ is written $\varepsilon_{(ijk\ldots)m}$, where m is the replication subscript.
2. For each term, classify subscripts: **live** (present, not in parentheses), **dead** (present, in parentheses), **absent** (in the model but not in that term). E.g. $(\tau\beta)_{ij}$: i, j live, k absent; $\varepsilon_{(ij)k}$: k live, i, j dead.
3. DF of a term $= \prod$(#levels of dead subscripts) $\times \prod$(#levels -1 of live subscripts). E.g. $(\tau\beta)_{ij}$: $(a-1)(b-1)$; $\varepsilon_{(ij)k}$: $ab(n-1)$.
4. If an interaction contains at least one random effect, the entire interaction is random.
5. Variance components: $\sigma^2_\tau, \sigma^2_\beta, \ldots$; fixed factors: $\sum \tau_i^2/(a-1)$, $\sum\sum (\tau\beta)_{ij}^2/((a-1)(b-1))$, …
6. Replicates are always random.

Steps (restricted model):

S1. Write the terms of the model as row labels; column headings are the subscripts, with F/R and the number of levels above them.

S2. In each row, write 1 in a column if the column's subscript is *dead* in that row's term.

S3. In each row, if the column subscript is *live* in the term, write 0 if the column factor is fixed (F) and 1 if random (R).

S4. Fill the remaining cells (absent subscripts) with the number of levels shown above the column.

S5. For the EMS of a term, cover the columns headed by the term's live subscripts.

S6. Consider every row that contains at least all the subscripts of the term; multiply the remaining numbers in that row and attach the row's fixed factor or variance component. Sum these.

Example 9.13: Two-factor mixed model via the algorithm (A fixed, B random)

	F	R	R	
	a	b	n	
Term	i	j	k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_{\tau\beta} + bn\sum \tau_i^2/(a-1)$
β_j	a	1	n	$\sigma^2 + an\sigma^2_{\beta}$
$(\tau\beta)_{ij}$	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta}$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

E.g. for β_j: cover column j; rows containing j: β_j (remaining $a\cdot n$), $(\tau\beta)_{ij}$ (remaining $0\cdot n = 0$), ε $(1\cdot 1)$. So $\mathrm{E}(MS_B) = an\sigma^2_\beta + 0 + \sigma^2$ — the zero in the fixed column is exactly the restriction $\sum_i (\tau\beta)_{ij} = 0$. For τ_i: cover i; rows containing i: τ_i (bn), $(\tau\beta)_{ij}$ $(1\cdot n)$, ε (1).
With A random and B fixed the table is $\tau_i : [1, b, n]$, $\beta_j : [a, 0, n]$, $(\tau\beta)_{ij} : [1, 0, n]$, $\varepsilon : [1, 1, 1]$, df $a-1$, $b-1$, $(a-1)(b-1)$, $ab(n-1)$.

Example 9.14: Three factors, A and C random, B fixed

$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$.

	R	F	R	R	
	a	b	c	n	
Term	i	j	k	l	EMS
τ_i	1	b	c	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + bcn\sigma^2_{\tau}$
β_j	a	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma} + acn\frac{\sum \beta_j^2}{b-1}$
γ_k	a	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	1	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$
$(\tau\gamma)_{ik}$	1	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + an\sigma^2_{\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	1	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

Derivation for β_j: cover column j; the rows containing j are β_j $(a\cdot c\cdot n)$, $(\tau\beta)_{ij}$ $(1\cdot c\cdot n)$, $(\beta\gamma)_{jk}$ $(a\cdot 1\cdot n)$, $(\tau\beta\gamma)_{ijk}$ $(1\cdot 1\cdot n)$, ε (1). For τ_i: rows with i: τ_i (bcn), $(\tau\beta)_{ij}$ $(0\cdot cn = 0)$, $(\tau\gamma)_{ik}$ $(b\cdot 1\cdot n)$, $(\tau\beta\gamma)$ (0), ε.
Test for $H_0 : \beta_j = 0$. No single mean square has expectation $\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma}$. But

$$\mathrm{E}(MS_{AB}) + \mathrm{E}(MS_{BC}) - \mathrm{E}(MS_{ABC}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma},$$

so use the approximate (Satterthwaite) test

$$F_0 = \frac{MS_B}{MS_{AB} + MS_{BC} - MS_{ABC}} \quad \text{or} \quad F_0' = \frac{MS_B + MS_{ABC}}{MS_{AB} + MS_{BC}},$$

with denominator df $\nu = \dfrac{(MS_{AB} + MS_{BC} - MS_{ABC})^2}{\frac{MS_{AB}^2}{f_{AB}} + \frac{MS_{BC}^2}{f_{BC}} + \frac{MS_{ABC}^2}{f_{ABC}}}$ (and for F_0' both numerator and denominator df are of this form).